

Handspring™ Lab session

Robert Gosselin
Developer Technical Support, Handspring,Inc.

3Com
More connected.

Agenda

PalmSource 99

- Differences between Visor and other platform
- Developing for Handspring's handheld computer
- Springboard™ expansion slot
  - Software API and considerations
- Lab:
  - Compile the SDK sample code
  - Flash a memory module

Visor Handheld

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- Differentiation from other Palm OS® platforms
  - New applications
    - CityTime
    - DateBook+
    - Advanced Calculator (MathLib is included in ROM)
  - USB connection to desktop (serial is optional)
  - Out-of-box support for PC and Mac
  - Integrated microphone
- Springboard expansion slot
  - Open mechanical architecture
  - Extension of the CPU bus

Developing for Handspring's Handheld Computer

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- What you need to know
  - Palm OS® 3.1-compatible platform with added support for
    - USB
    - Springboard expansion slot
    - Euro $ (€) character
  - True plug & play implementation
    - Some special considerations...
- Designed for hardware expandability
- Can also be used for software distribution
  - Recommended vendor for manufacturing application-only modules

Developing for Handspring's Handheld Computer *(continued)*

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- You also need to know about Handspring's API extensions, but only if you are developing Springboard module

• HsAppEventHandlerSet	Register event handler procedure
• HsAppEventPost	Enable event posting by application event handler
• HsCardAttrGet	Retrieve attribute
• HsCardAttrSet	Set attribute
• HsCardErrTry & HsCardErrCatch	Enable safe recovery for module removals
• HsCardEventPost	Queue an event
• HsCardPatchInstall	Enable patch installs
• HsCardPatchPrevProc	Get address of previous system trap implementation
• HsCardPatchRemove	Remove setup application patch
• HsDatabaseCopy	Copy Palm OS® database
• HsEvtResetAutoOffTimer	Reset auto-off timer

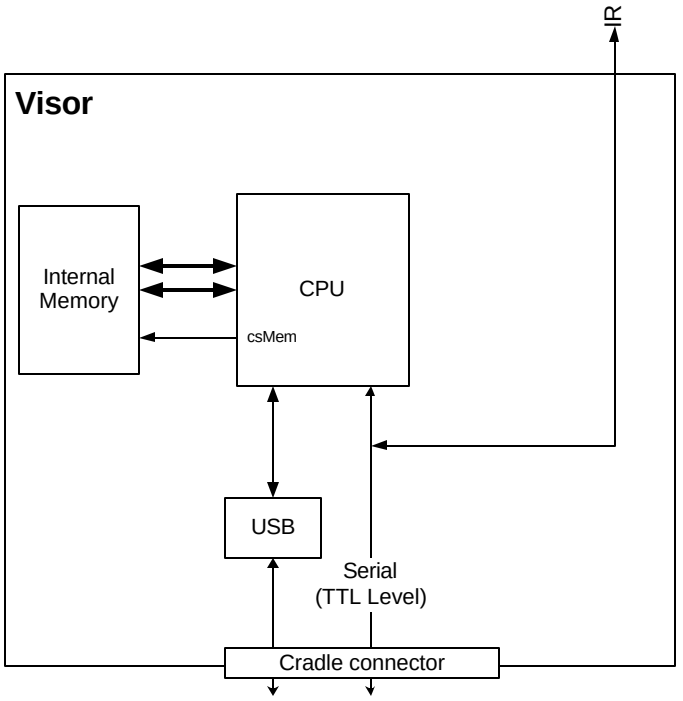
Details are available on our website in the Handspring Development Kit manual

Developing for Handspring's Handheld Computer *(continued)*

- If you are simply doing application development and you want to publish on a module, use a recommended vendor that manufactures application-only module (Smart Modular)
  - Simply use the Palm™ device makeROM utility to create a ROM image and send the file to them ⇒ they will manufacture module for you!

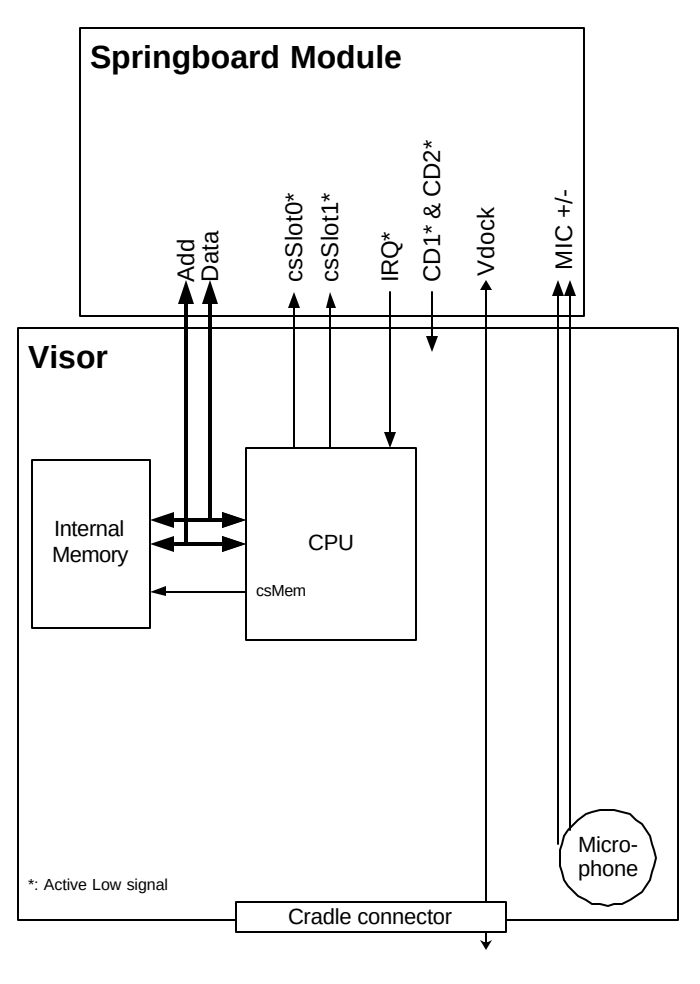
Visor Hardware Details

- DragonBall-EZ CPU (16.58 MHz)
- Combined USB & serial port
- Standard Palm OS® infrared
- Microphone wired directly to the Springboard expansion slot
- LCD
  - 160 x160 resolution
  - 2.2” x 2.2” (3.1” diagonal)
- Internal memory
  - 2 MB or 8 MB
- Springboard expansion slot memory
  - Maximum 32 MB



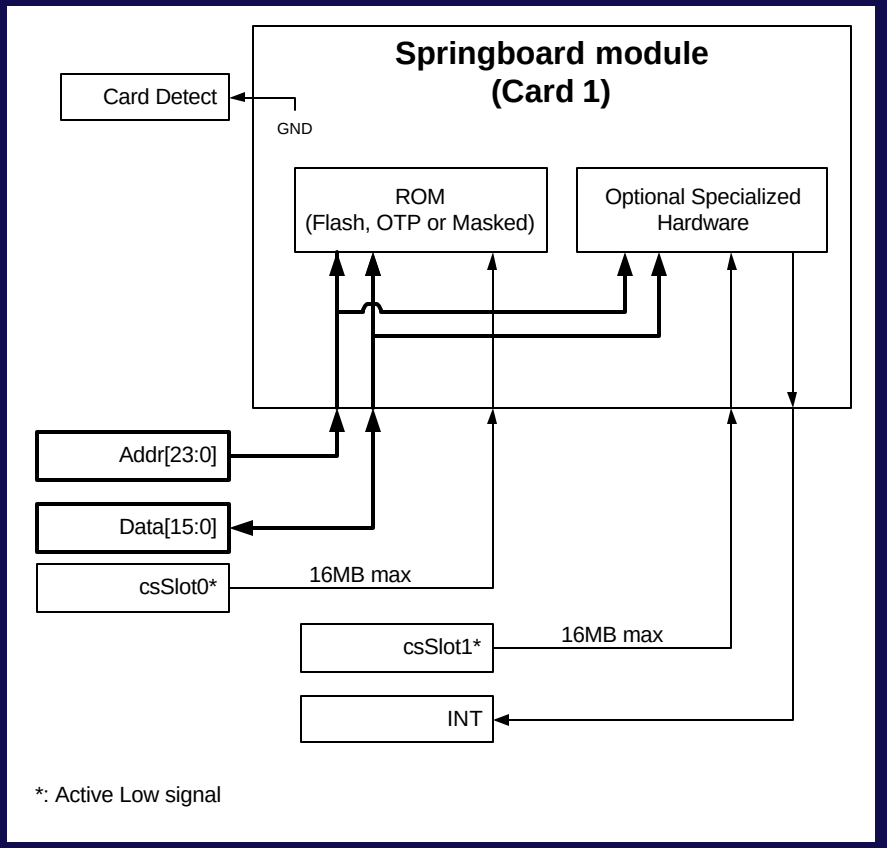
Springboard Interface Details

- Extension of the CPU bus (memory-mapped)
  - 24 address lines
  - 16-bit data bus
  - Control signals
    - CS1*, CS2*, OE*, WE*, Int*, Reset*
  - Card detection
    - CD1*, CD2*
  - Others
    - VDOCK (module charging power)
    - Mic+, Mic-
  - Power
    - VCC: 100 mA maximum @ 3.3V
    - GND
- ➔ *Check your power consumption!*



Springboard Module Components

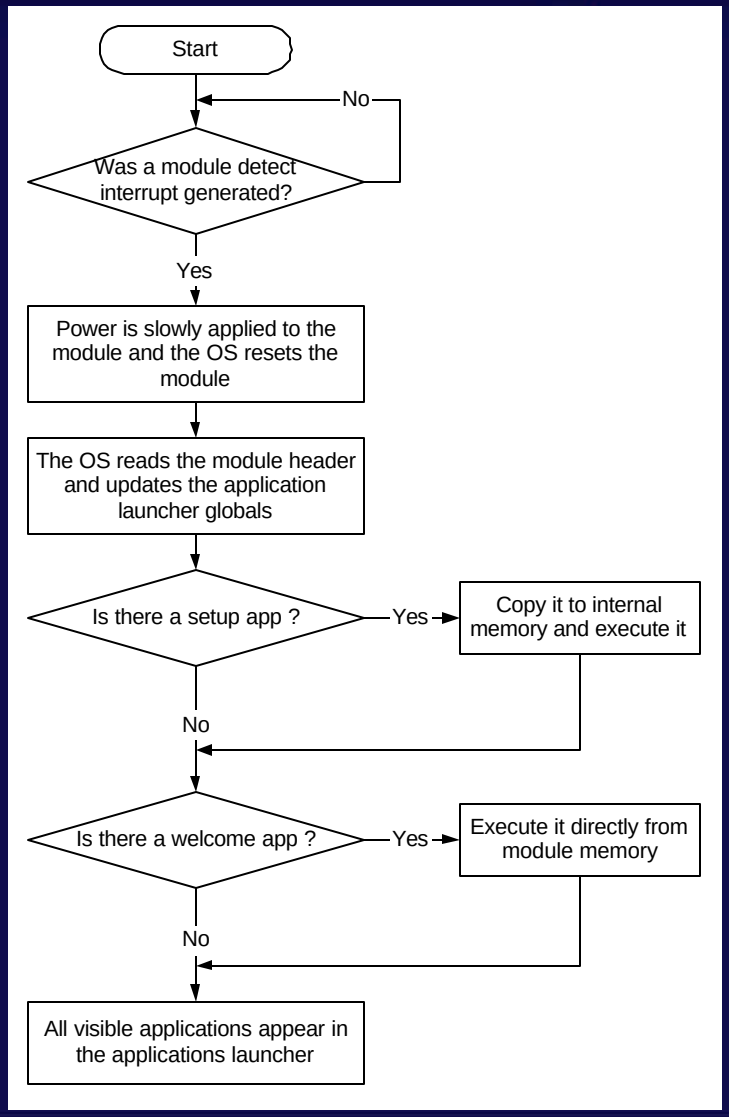
- ROM
  - Contains module header information
  - Any other applications needed for the module
  - Created with “Palm™ device makeROM” utility included in the SDK
  - Connected to csSlot0*
- Any special hardware
  - Connected to csSlot1*



Inserting a Springboard Module

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- What happens?
  - Interrupt is sent to the CPU
  - H/W powers-up the module slowly
  - OS maps the chip select signals to the default address range
  - OS resets the module
  - OS checks the card header; if valid, updates globals
  - If “setup app” is present, copy it to internal memory and execute it
  - When finished executing the “setup app”, if “welcome app” is present, execute it from module memory
  - Return to Launcher

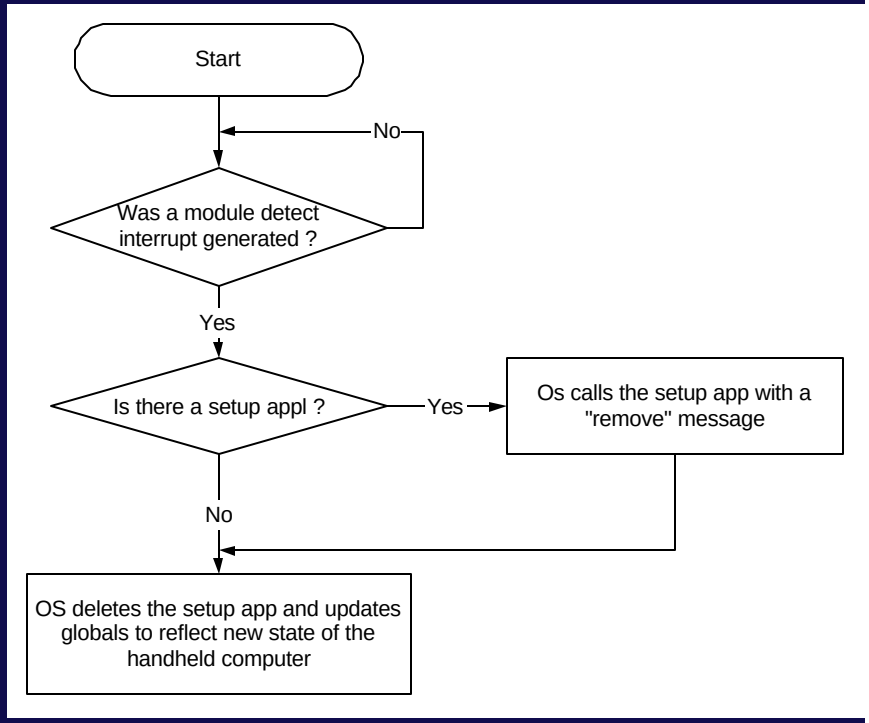


Inserting a Springboard Module *(continued)*

- If a module is already plugged in during a device reset, all applications and panels on the module are sent the standard Palm OS® reset action code (`sysAppLaunchCmdSystemReset`), as are all normal built-in applications and panels.
- When a module is hot-inserted at run time, the system also broadcasts the reset action code, but only to applications (not panels) on the module that have the `dmHdrAttrResetAfterInstall` bit set in their database header.
  - Ensures that applications that require the reset notification are notified
  - Does not slow down the insertion process

Removing a Springboard Module

- What happens?
  - Interrupt is sent to the CPU or a bus error occurs
  - OS “de-maps” chip select signals
  - If “setup app” is present, OS executes it with a “removal” message
  - OS deletes the “setup app” and updates globals
  - Return to Launcher (if needed)



Removing a Springboard Module *(continued)*

- Application is accessing or running off the module when it is removed?
  - If it is a “normal” application:

    ⇒ The OS takes care of it.

  - If it is a “non-application”, like interrupt handler:

    ⇒ The OS does not really know how to deal with it.

    ⇒ You must use HsCardErrTry/HsCardErrCatch macros.

Other Things to Know

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- How to check if this is a Handspring device:

```
DWord value;
err = FtrGet (hsFtrCreator, hsFtrIDVersion, &value);
if (!err)
      {
      // Since FtrGet() did not return an error, we can
      // safely make Handspring specific API calls like
      // HsCardAttrGet(), HsCardAttrSet(), etc.
      }
```

- Mechanism for copy protection:

```
char cardName[32];
Err err;
err = MemCardInfo (1 /*cardNo*/, cardName, 0 /*manufName*/,0 /*versionP*/,
     0 /*crDateP*/, 0 /*romSizeP*/, 0 /*ramSizeP*/, 0 /*freeBytesP*/);
if (err || StrCompare (cardName, "MyCardName"))
      {
      DisplayCopyProtectError();
      }
```

Setup Application

- Its creator ID must be “HsCd”
- Setup App receives the following information when module is
  - inserted:
    - launch code: `hsSysAppLaunchCmdInstall`
    - parameter block contains an `isReset` field.
  - removed:
    - launch code: `hsSysAppLaunchCmdRemove`
- Note:
  - Can’t use application global or static variables; all variables must be normal stack- or register-based local (same as when an app is called with Find)
  - Since most module software requires globals of some kind. The hsCardAttrCardParam attribute of the module is provided for this purpose. Use this attribute to store a 32-bit pointer to the module's globals; it can be set using HsCardAttrSet() and retrieved using HsAppEventHandlerSet(). In addition, this attribute is automatically passed to the module's interrupt handler as a parameter on the stack.
  - When a module is removed the “setup app” cannot display any alerts (using FrmAlert() or similar mechanism) or make system calls that might display UI elements or process user interface events.

Welcome Application

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- Welcome application:
  - Its database name must be “CardWelcome”
  - To have it automatically launched during a reset (even if the module is already inserted) you must use the HsWR token when building your binary image with Palm™ device makeROM (“`-tokStr HsWR 1`”):

Recommended Application Development Process

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- Design your application as before
- Test it in internal memory
- Test it in module memory using an 8MB Flash Module and its “File Mover” app to transfer your application on the module.
- If you want to sell it on a module:
  - Create a ROM image using Palm™ device MakeROM
    - flash the memory module using Palm-Debugger
    - send your image to recommended vendor
  - Recommended vendor for application-only module:
    - Smart Modular, Dave Floriani, 510-624-8137, dave.floriani@smartm.com, www.smartm.com

Recommended Development Process…

- Develop your application using either GNU or CodeWarrior for Palm OS® technology
  - If you are not using any Handspring API calls, you can debug your application using POSE with the Palm V™ organizer ROM
  - After you have finished debugging your application or if you are using Handspring API calls, do your debugging on the Visor itself
    - direct support (USB or serial) with the GNU tool kit and Palm Debugger on the PC.
    - Need serial cradle if using CodeWarrior on the PC or the MAC:

      when doing “Shortcut .. 2”, press and hold the up button on the device until you see the source code window

Application Development Considerations

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- When your application is running on a Springboard-compatible module
  - The handheld's internal memory is defined as Card 0
  - The Springboard module's memory is on Card 1
    - Opening & closing databases…
    - or when using any calls that has the parameter "CardNo"
  - The memory on the module could be read-only
    - Flash module
    - OTP (One-Time Programmable) module
    - Masked ROM module
      - API calls like MemPtrRecoverHandle() don't work

Application Development Considerations *(continued)*

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- If you are saving status information or preferences
  - The preferred way to save status information or preferences is to use the standard Palm OS® call PrefSetAppPreferences()
  - If you must use your own database, follow these steps
    - Try writing to the card # your application is running on
    - If it fails, try writing to Card 0

Application Development Considerations *(continued)*

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- How to handle the removal of a Springboard module while your application is running
  - Write a “valid bit” as a last step when saving a record
  - When opening a record, verify that it’s valid (if it’s a database, make sure that it works). If it isn’t valid, don’t crash, simply advise the user if necessary...
  - Try to save your “preferences” or status information as often as needed, especially after changes!

Application Development Considerations *(continued)*

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- Design your application for potential updates:
  - Check for the presence of newer version of library or databases before actually running, this will enable updates w/o hardware modification.

What's Available Today

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- Recommended vendors for module parts
  - Springboard module plastics: 5 versions!
  - Connectors for modules or bottom (docking) accessories
- Recommended vendor for building application-only modules
- Complete development kit
  - Mechanical specifications
  - Electrical specifications
  - Springboard-compatible logo
  - GNU tools: compiler & linker
  - Palm debugger for source level debugging on the PC
  - Visor support in Metrowerks CodeWarrior for Palm OS® technology

What's Available Today *(continued)*

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- 8 MB Flash Module for application development
- Two subscription programs for developers
  - No-cost program
    - E-mail support for technical questions, but no guaranteed response time
    - E-mail support for marketing questions
    - Access to Handspring's Knowledgebase, including FAQs and technical notes
    - Opportunities to take advantage of Handspring's marketing programs, such as trade show participation, press release support and inclusion in Handspring collateral that will highlight developer products
  - Premier program (3000$/year)
    - Everything in the No-Cost program plus:
      + Guaranteed response time (1 business day)
      + 20% discount on the purchase of Visor and/or Visor Deluxe, with a maximum of 20 total units per year
      + 20% discount on the purchase of select Handspring accessories

What's Coming Soon

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- Diagnostic Module (Hardware Reference Design)
  - Flash memory
  - Uses the "Battery Module" plastics
  - Test register for Read/Write
  - Sample code (already in the SDK)
  - UART (16C560)
  - LED
  - Interrupt button
- Palm OS® Emulator support for Visor
- Handspring module for the Palm OS® class
- Training module on how to use the tools
- More sample code and documentation
  - For example, an SDK for the USB port
- Sync-up forum to discuss issues & new features
- Palm™ device makeROM for the MAC

LAB

- GNU tool kit (PC) and CodeWarrior (PC and MAC)
  - Compile sample application available in our SDK
  - Do source level debugging
- GNU tool kit (PC only)
  - Create a ROM Image using Palm™ device makeROM
  - Flash a 8MB memory module

LAB *(continued)*

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- Compile the sample code:
  - Tex2Hex:
    - standard way
    - As a welcome application
- Transfer your app to Visor, test it.
- Move your app to the 8MB module and test it.

LAB *(continued)*

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- Let's flash a module:

```
Palm-MakeROM   -op create                    \
-hdr 0x08000000                              \
-chName "SampleROMCard"                      \
-chManuf "Handspring, Inc."                  \
-chVersion 0x0100                            \
-romName "ROM Store"                         \
-romBlock 0x08000000 0x00010000              \
-chRomTokens 0x0800FF00 0x0100               \
-tokStr HsAT 200                             \
-romDB ”Tex2Hex.prc"                         \
-romDB "MyApp.prc"                           \
-o CardROM.bin
```

Conclusions

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- If you know how to program for Palm OS® technology, you know how to program for Visor
- All the information you need is on the Web
- Subscribe to one of the developer programs – it will help you and us
- For technical questions
  - On the Springboard expansion slot and/or Handspring's API extensions
    - Use the Web form ⇒ you will get your answer faster
    - Send an e-mail to DevSupport@handspring.com
  - On Palm OS® technology
    - Use the standard Palm Developer Technical Support Group and their website

All information @ www.handspring.com


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